

Parametric Survival Models Report

METIMMOX-1 Health Economic Evaluation

Ben Geisler

2025-11-03

Table of contents

1	Overview	2
2	Population Characteristics including Biomarker Prevalences	2
3	Model Selection and Comparison	2
3.1	Analysis Specifications	2
3.2	Data Considerations	2
3.3	Model Validation and Goodness of Fit	3
4	Model Comparison	3
5	Model Coefficients	3
5.1	Overall Survival (OS) Model Coefficients	3
5.2	Progression-Free Survival (PFS) Model Coefficients	4
6	Population Survival Curves	5
7	Biomarker-Specific Results	9
8	Summary	21

1 Overview

This report presents all parametric survival models fitted for the METIMMOX-1 health economic evaluation. The models compare different biomarker strategies for overall survival (OS) and progression-free survival (PFS) outcomes. The parametric survival models were fitted to compare biomarker-guided immunotherapy selection strategies in metastatic microsatellite stable/proficient mismatch repair (MSS/pMMR) colorectal cancer patients.

2 Population Characteristics including Biomarker Prevalences

****Continuous variables:****

- Age (mean): 63.98 years

****Categorical variables (most frequent category used):****

- Sex: 1 (0=female, 1=male)

- CRP: 0 (0=negative, 1=positive)

- TLR: 1 (0=negative, 1=positive)

- TMB_BRAF: 0 (0=negative, 1=positive)

****Biomarker prevalences in population:****

- CRP positive: 33.8 %

- TLR positive: 64.7 %

- TMB_BRAF positive: 44.9 %

****Data completeness:****

- Complete cases used: 65 of 74 observations (87.8 %)

3 Model Selection and Comparison

3.1 Analysis Specifications

- **Distributions tested:**

- Exponential

- Weibull

- Weibull PH

- Gompertz

- Log-logistic

- Log-normal

- Generalized Gamma

- Gamma

- Generalized F Distribution

- **Covariate structures:** Full models (age, sex, treatment, biomarkers) vs models without age/sex adjustment

3.2 Data Considerations

- **Complete case analysis:** Only patients with complete covariate data included

- **Reference patient:** Modal categories for categorical variables, mean for continuous variables

- **Time horizon:** Models fitted to observed follow-up period with extrapolation capability

- **Biomarker prevalences:** Taken from the METIMMOX-1 trial and may or may not reflect real world settings

3.3 Model Validation and Goodness of Fit

- **Selection criterion:** AIC is more appropriate here because:
 - BIC penalizes model complexity more heavily (uses $\log(n)$ vs 2 for AIC)
 - For prediction purposes, we want the model that best balances fit and complexity
 - BIC tends to select overly simple models that might miss important biomarker interactions
 - AIC is better for predictive modeling while BIC is better for identifying the “true” model
- **Validation:** Visual comparison with Kaplan-Meier curves and clinical plausibility assessment

4 Model Comparison

Table 1: Overall Survival (OS) model comparison using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	622.56	646.47	0.00	1.0	***
Full Model	weibull	623.44	647.36	0.88	2.5	
Full Model	weibullph	623.44	647.36	0.88	2.5	
Full Model	llogis	623.68	647.60	1.12	4.0	
Full Model	lognormal	624.09	648.01	1.53	5.0	
Full Model	gengamma	624.56	650.65	2.00	6.0	
Full Model	genf	626.56	654.82	4.00	7.0	
Full Model	gompertz	633.09	657.01	10.53	8.0	
Full Model	exponential	654.14	675.88	31.58	9.0	

Table 2: Progression-Free Survival (PFS) model comparison using AIC and BIC

Structure	Distribution	AIC	BIC	Delta AIC	Rank	Selected
Full Model	gamma	453.43	477.35	0.00	1.0	***
Full Model	lognormal	453.46	477.38	0.03	2.0	
Full Model	weibull	454.19	478.11	0.76	3.5	
Full Model	weibullph	454.19	478.11	0.76	3.5	
Full Model	llogis	455.16	479.08	1.73	5.0	
Full Model	gengamma	455.28	481.37	1.85	6.0	
Full Model	genf	457.28	485.55	3.85	7.0	
Full Model	gompertz	467.57	491.49	14.14	8.0	
Full Model	exponential	483.52	505.26	30.09	9.0	

5 Model Coefficients

5.1 Overall Survival (OS) Model Coefficients

****Model specification:****

- Distribution: gamma
- Structure: full
- AIC: 622.56

Table 3: Regression coefficients for the selected OS parametric survival model

Parameter	Estimate	Lower 95% CI	Upper 95% CI	SE
shape	1.2275	0.8707	1.5843	0.1820
rate	-2.8975	-4.1041	-1.6908	0.6157
Age	0.0015	-0.0145	0.0175	0.0082
sex1	0.1454	-0.1450	0.4358	0.1482
RxExperimental arm	-0.2309	-0.7898	0.3281	0.2852
RxControl arm:crp1	-0.3570	-0.9767	0.2628	0.3162
RxExperimental arm:crp1	-0.3395	-0.7834	0.1044	0.2265
RxControl arm:tlr1	-0.8039	-1.2956	-0.3121	0.2509
RxExperimental arm:tlr1	-0.4109	-0.8282	0.0064	0.2129
RxControl arm:tmb_braf1	-0.0866	-0.5197	0.3465	0.2210
RxExperimental arm:tmb_braf1	-0.1859	-0.5920	0.2202	0.2072

5.2 Progression-Free Survival (PFS) Model Coefficients

****Model specification:****

- Distribution: gamma
- Structure: full
- AIC: 453.43

Table 4: Regression coefficients for the selected PFS parametric survival model

Parameter	Estimate	Lower 95% CI	Upper 95% CI	SE
shape	1.2426	0.8701	1.6152	0.1901
rate	-1.8647	-3.1036	-0.6258	0.6321
Age	-0.0009	-0.0173	0.0155	0.0084
sex1	0.1321	-0.1919	0.4562	0.1653
RxExperimental arm	0.4218	-0.1804	1.0240	0.3073
RxControl arm:crp1	0.0654	-0.7681	0.8990	0.4253
RxExperimental arm:crp1	-0.5754	-1.0481	-0.1026	0.2412
RxControl arm:tlr1	-0.9764	-1.5091	-0.4436	0.2718
RxExperimental arm:tlr1	-1.1914	-1.6095	-0.7732	0.2133
RxControl arm:tmb_braf1	-0.1055	-0.6027	0.3917	0.2537
RxExperimental arm:tmb_braf1	-0.4891	-0.9447	-0.0334	0.2325

6 Population Survival Curves

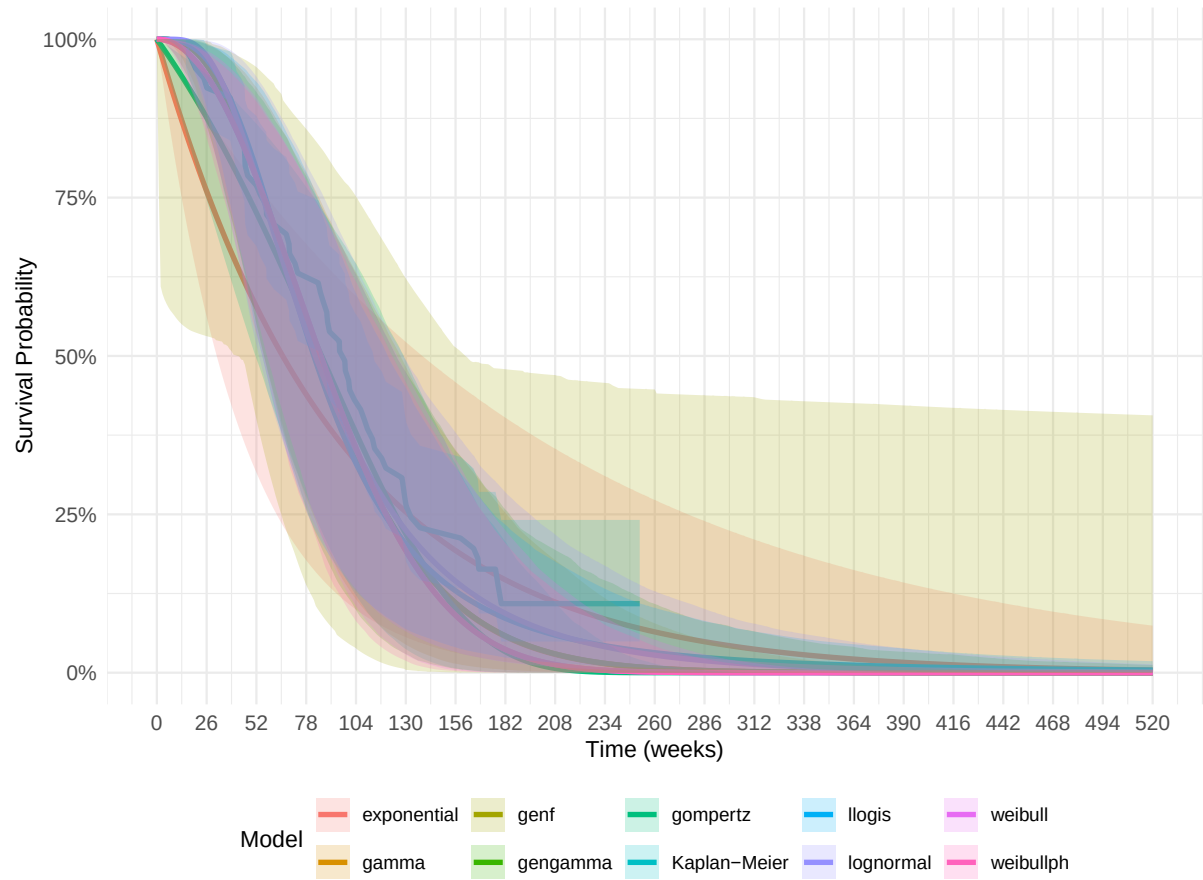


Figure 1: Overall survival curves for full models compared to Kaplan-Meier estimates

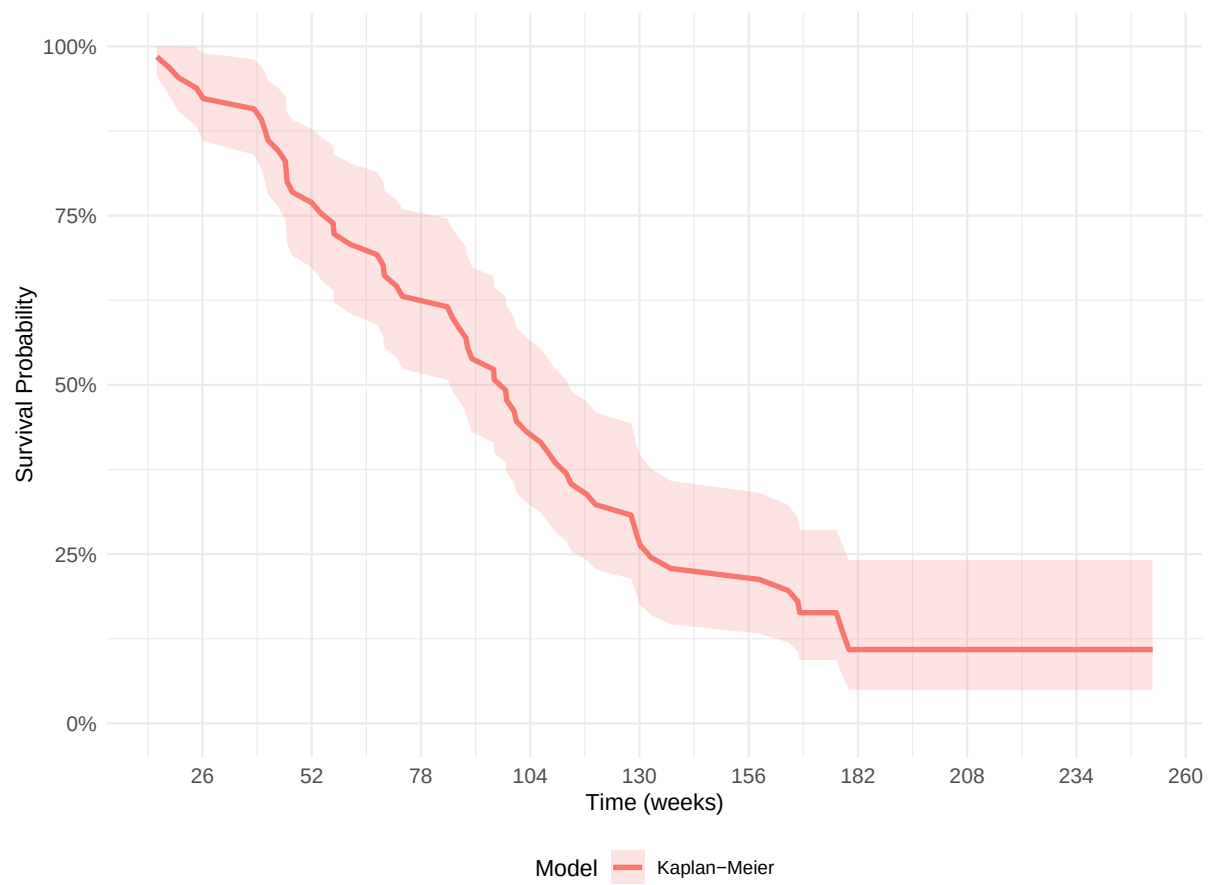


Figure 2: Overall survival curves for models without age/sex adjustment compared to Kaplan-Meier estimates

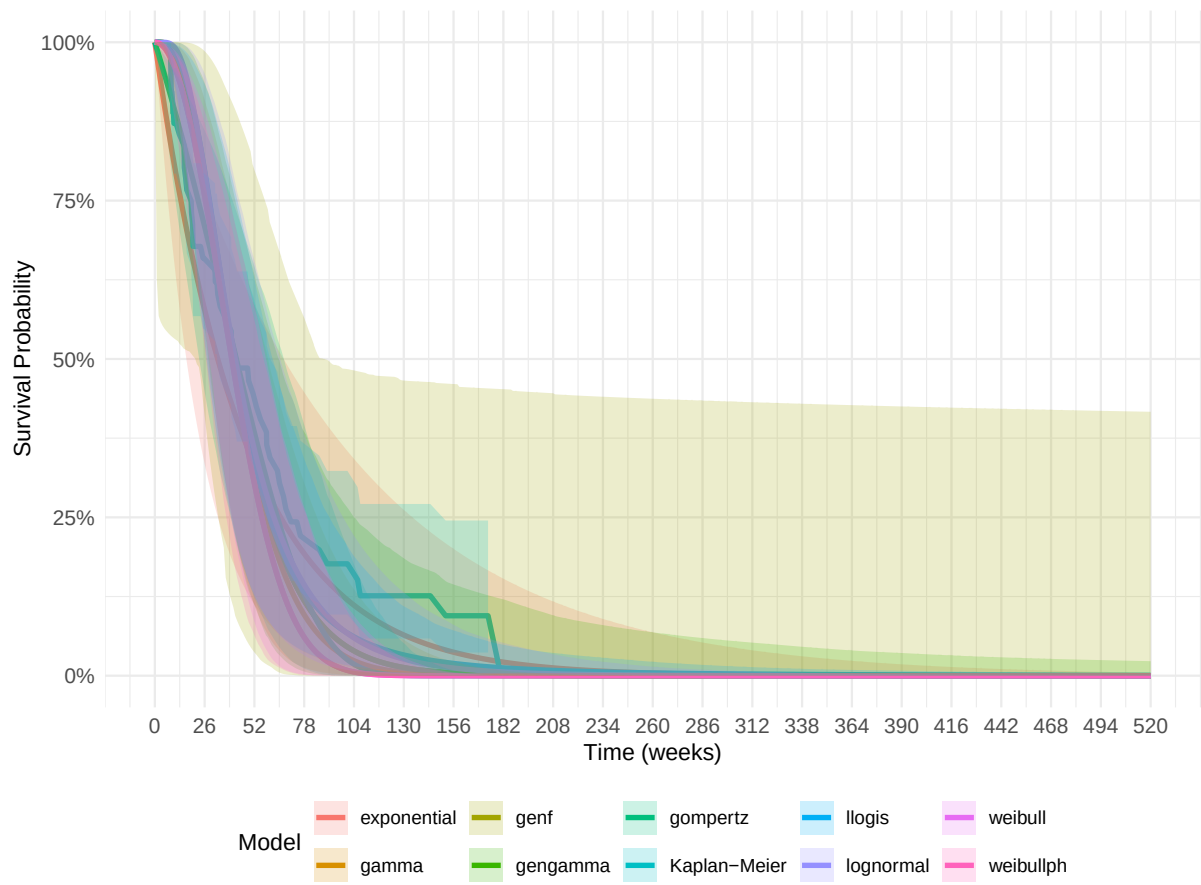


Figure 3: Progression-free survival curves for full models compared to Kaplan-Meier estimates

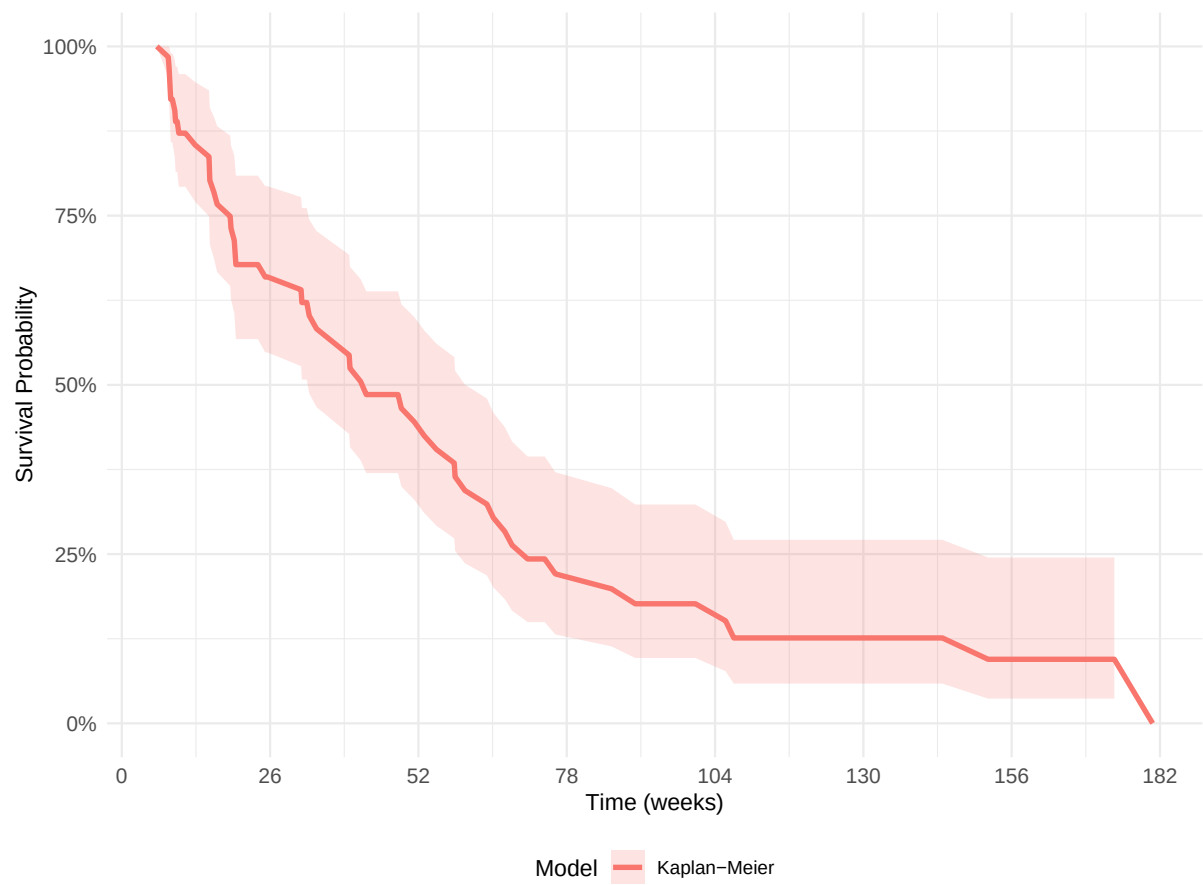


Figure 4: Progression-free survival curves for models without age/sex adjustment compared to Kaplan-Meier estimates

7 Biomarker-Specific Results

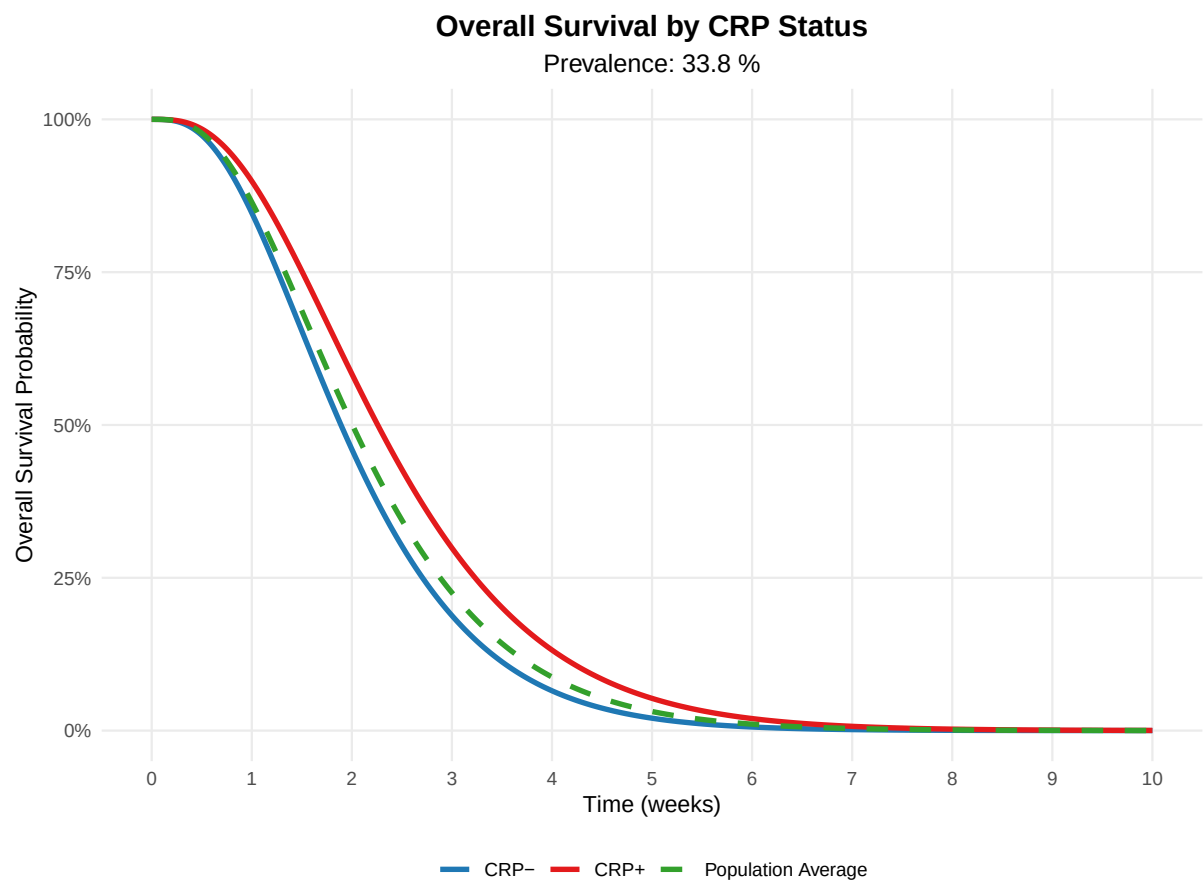


Figure 5: Overall survival for CRP biomarker-positive versus biomarker-negative patients

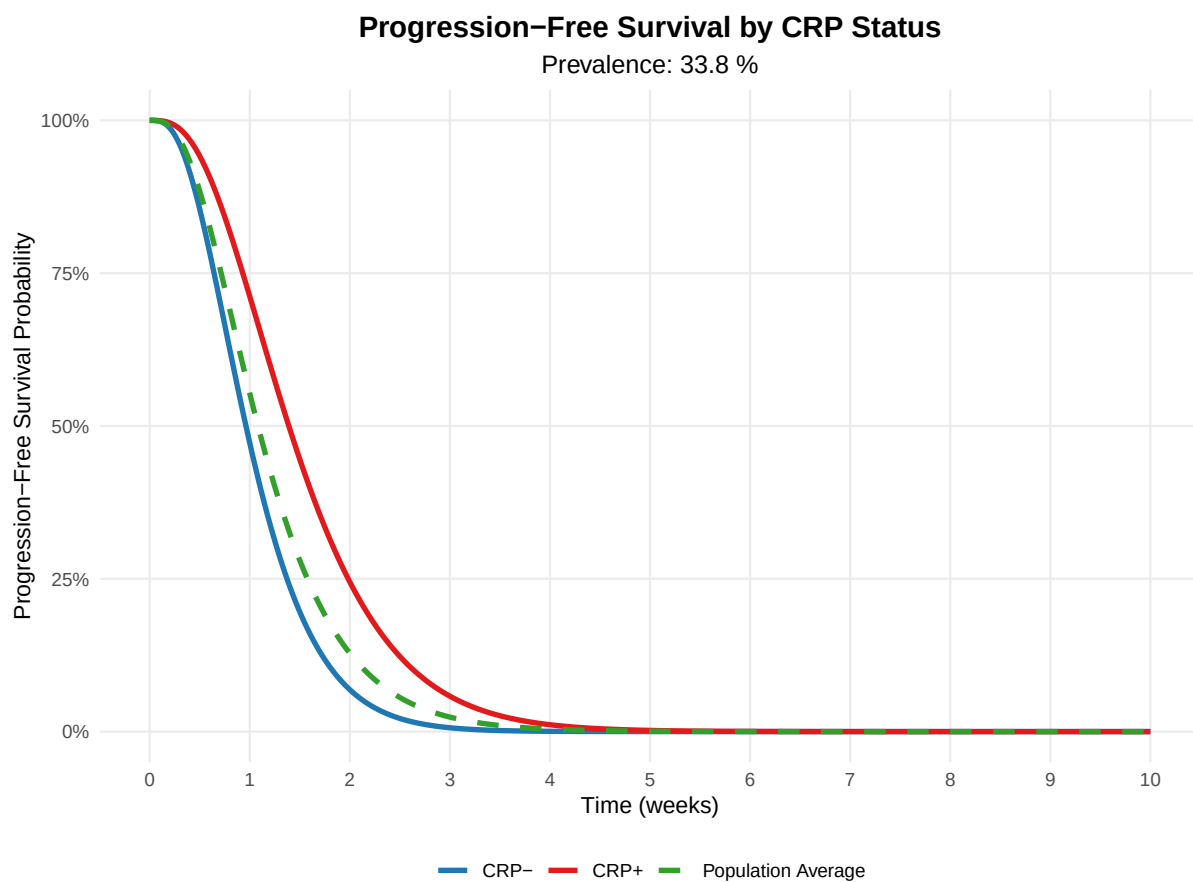


Figure 6: Overall survival for CRP biomarker-positive versus biomarker-negative patients

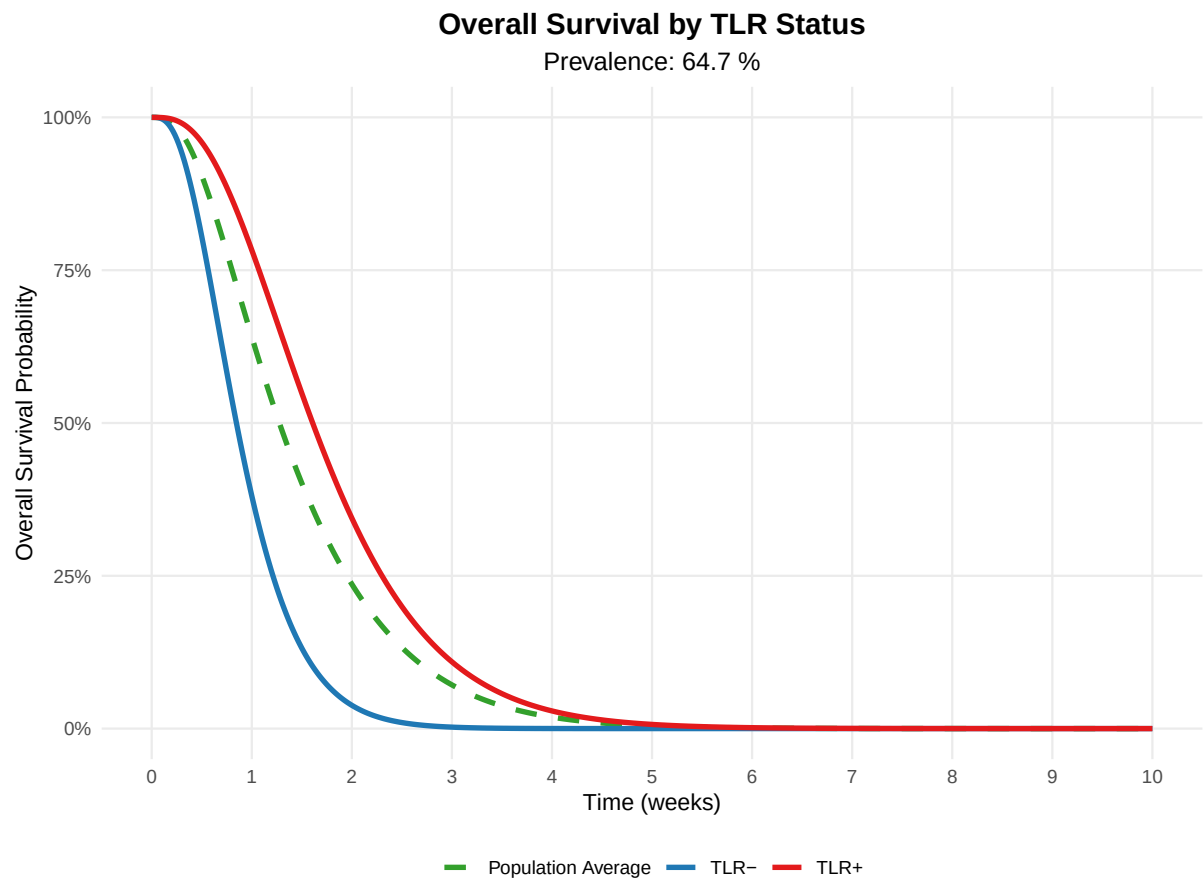


Figure 7: Overall survival for CRP biomarker-positive versus biomarker-negative patients

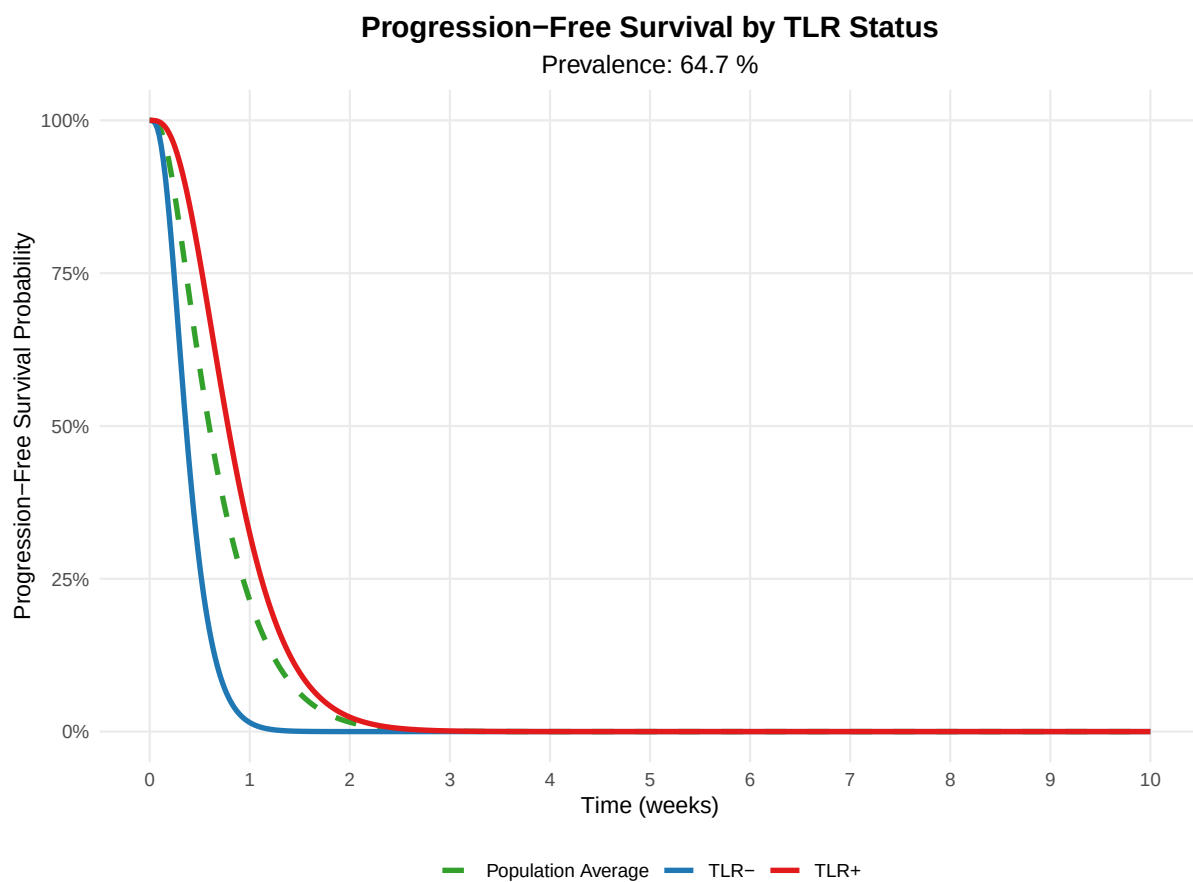


Figure 8: Overall survival for CRP biomarker-positive versus biomarker-negative patients

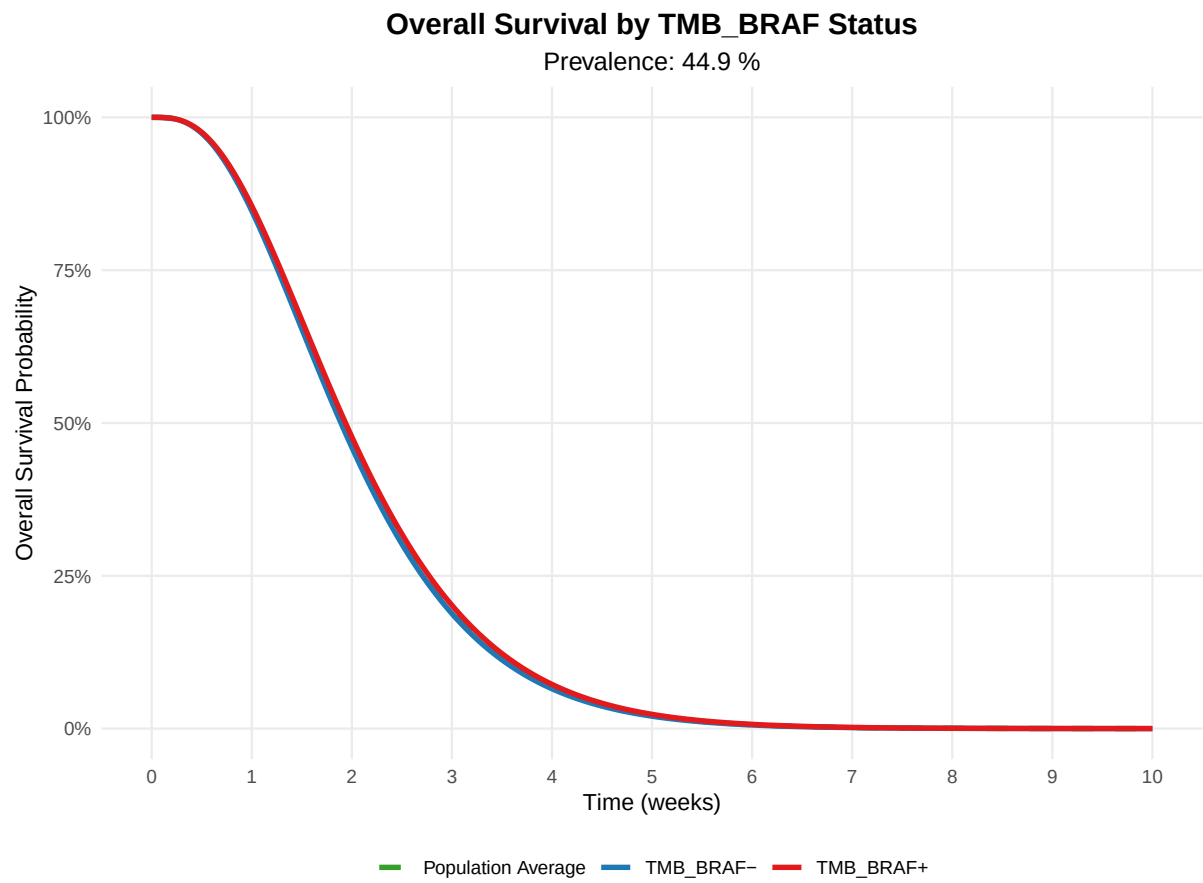


Figure 9: Overall survival for CRP biomarker-positive versus biomarker-negative patients

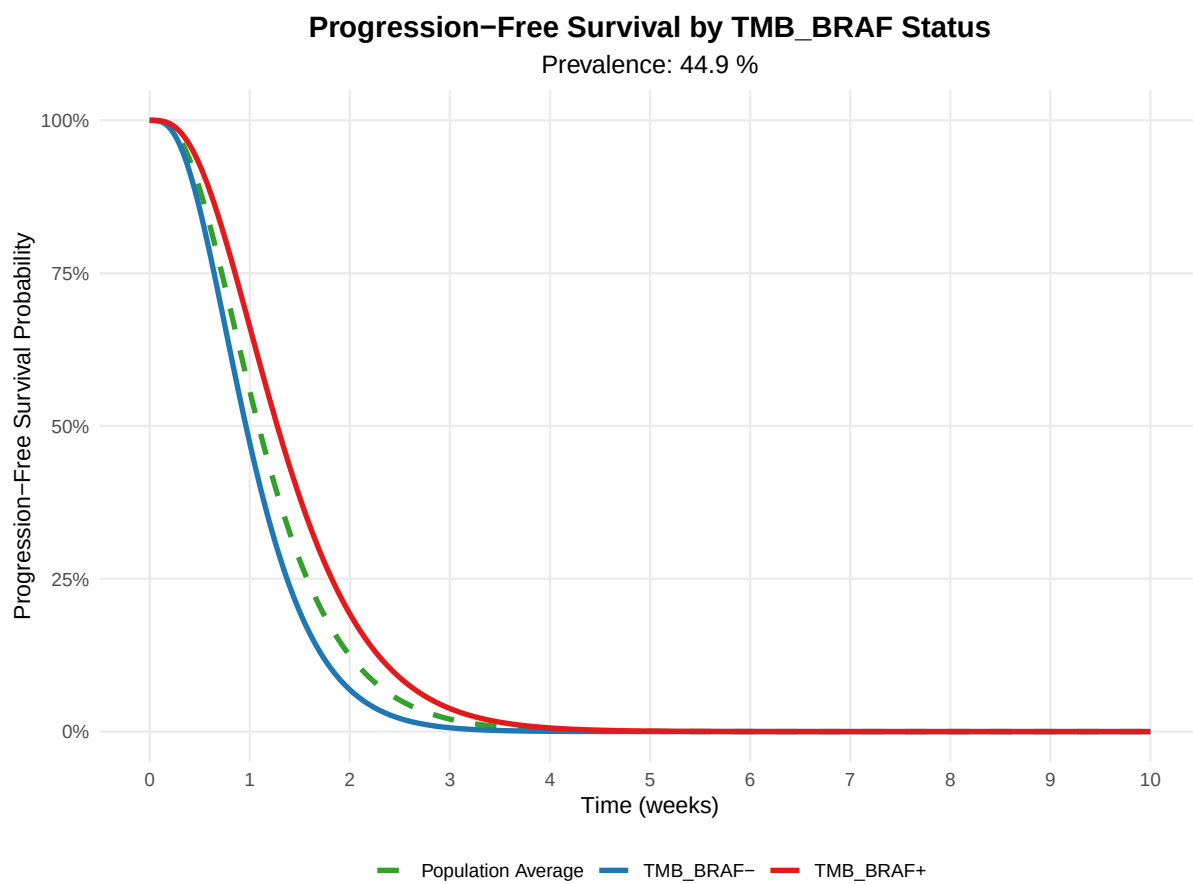


Figure 10: Overall survival for CRP biomarker-positive versus biomarker-negative patients

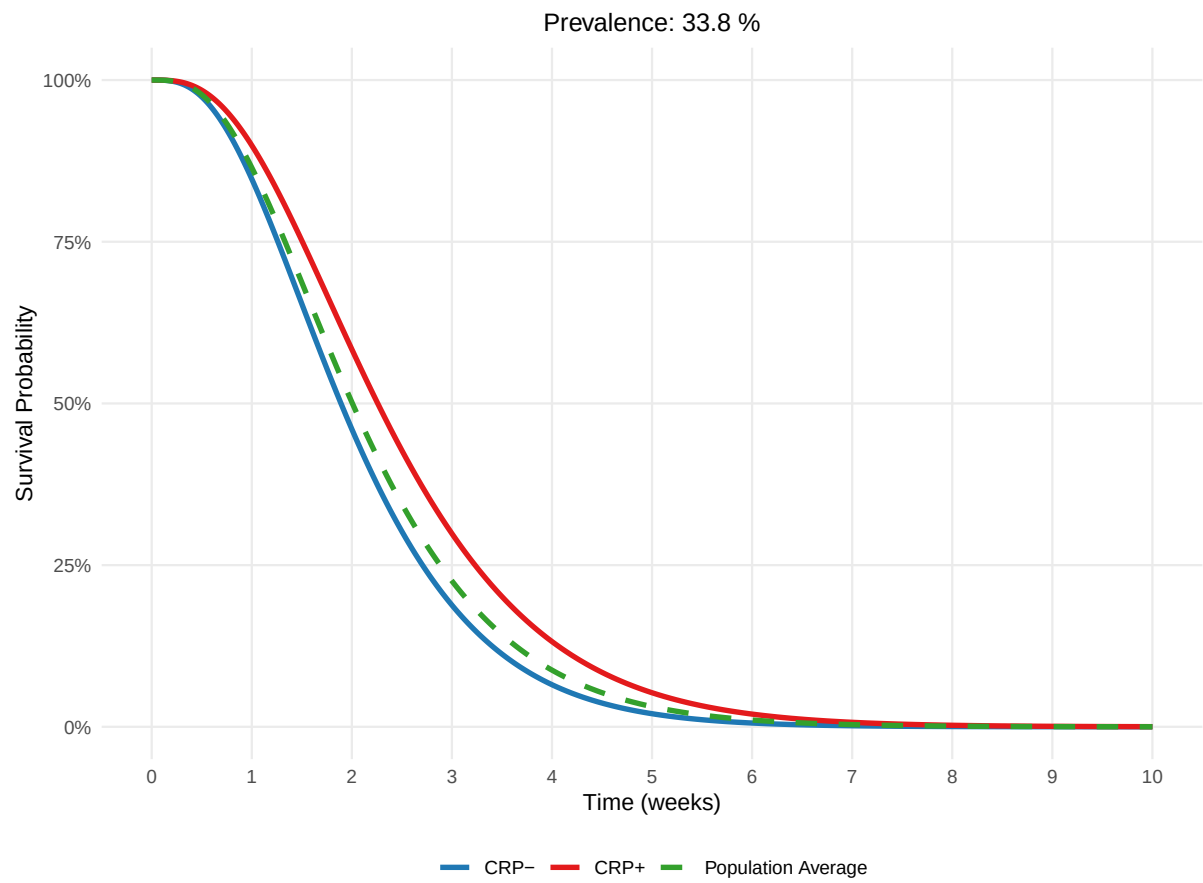


Figure 11: Overall survival for CRP biomarker-positive versus biomarker-negative patients

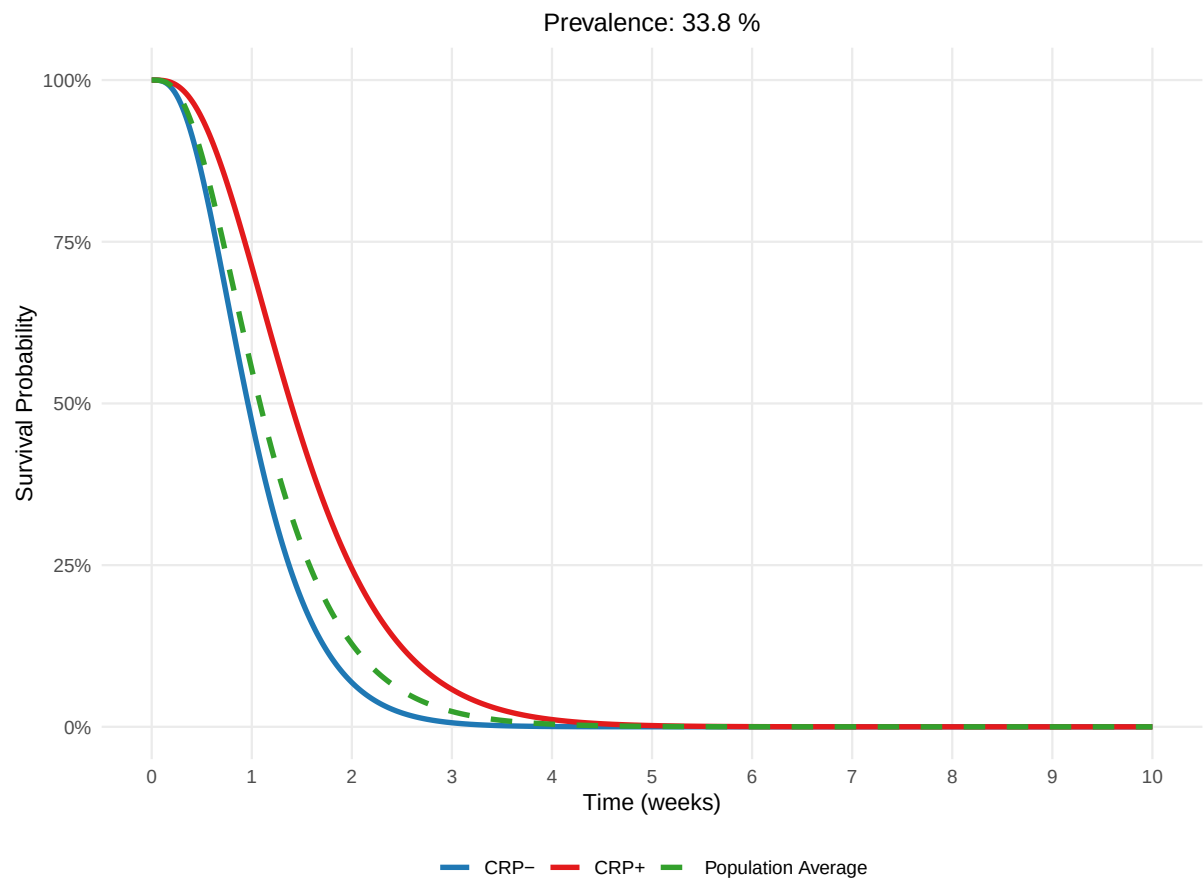


Figure 12: Progression-free survival for CRP biomarker-positive versus biomarker-negative patients

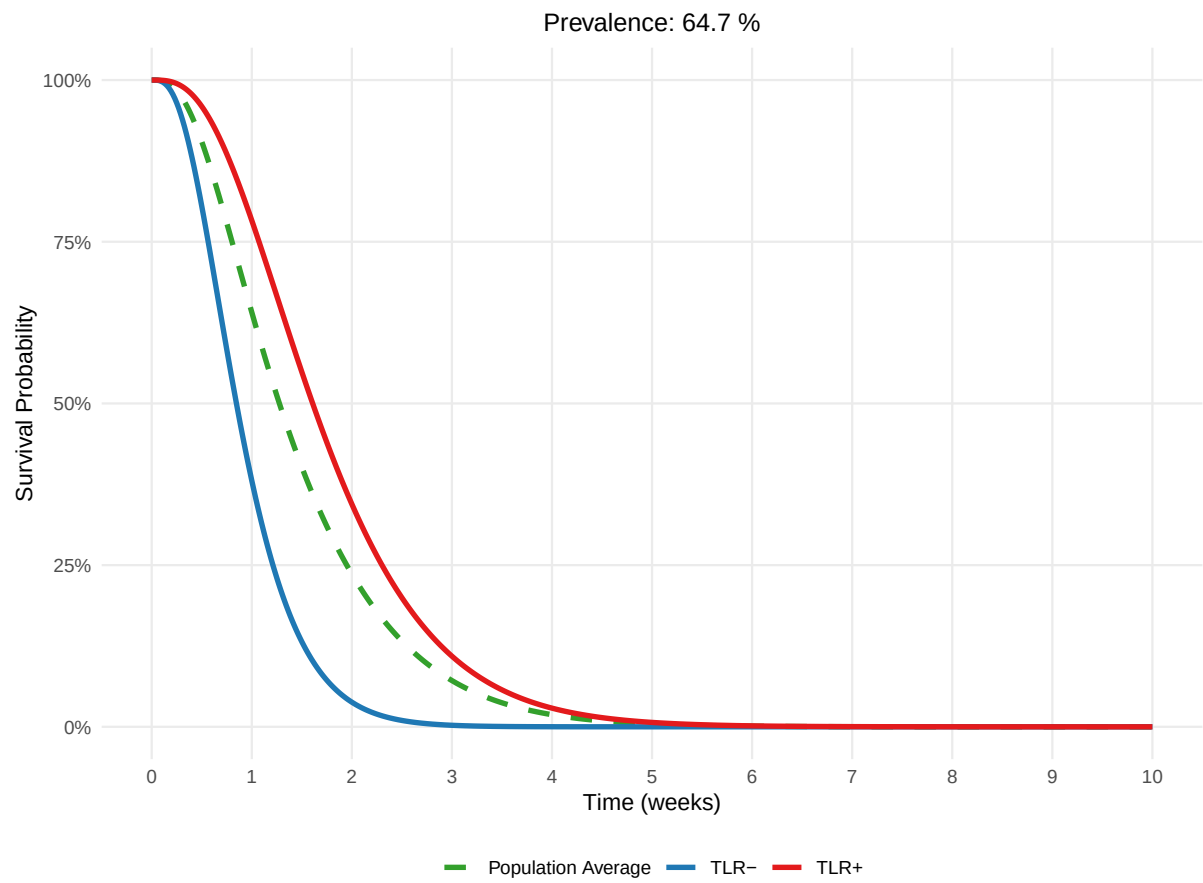


Figure 13: Overall survival for TLR biomarker-positive versus biomarker-negative patients

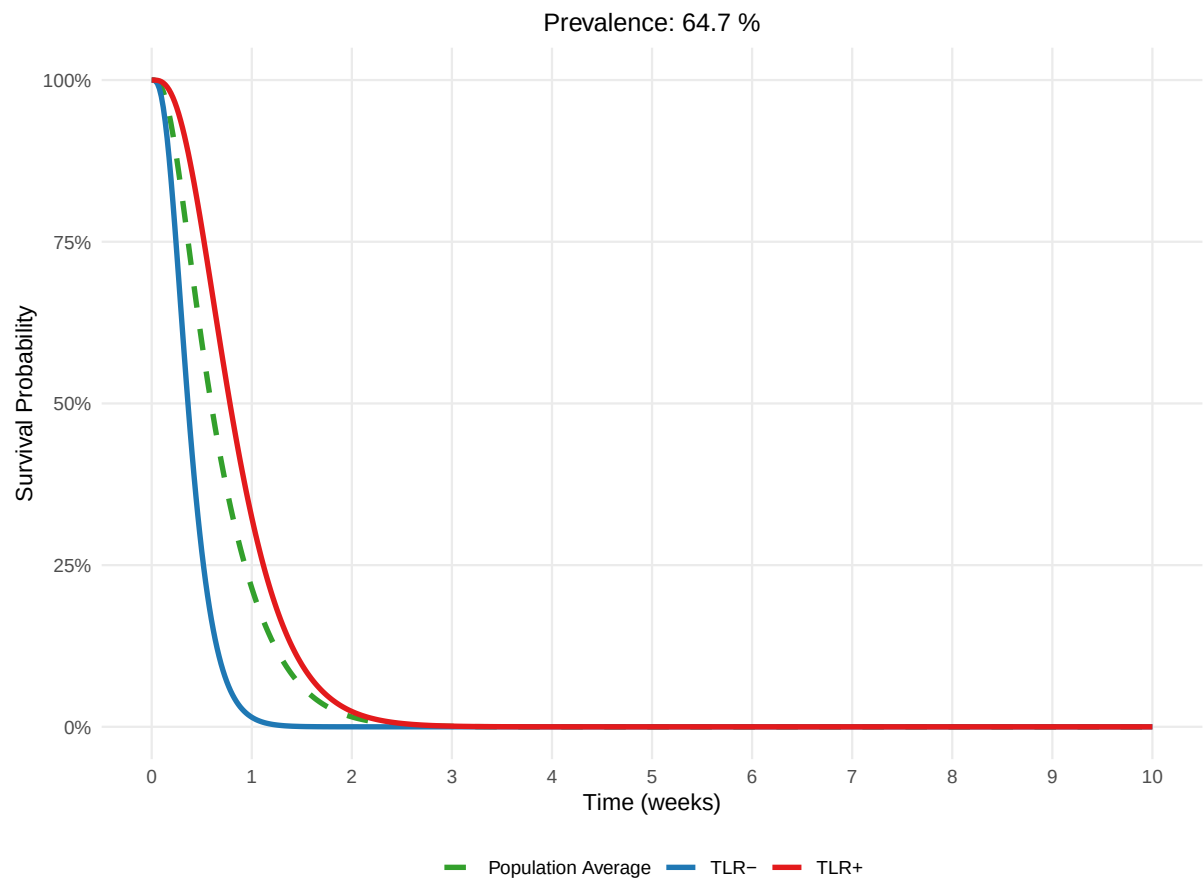


Figure 14: Progression-free survival for TLR biomarker-positive versus biomarker-negative patients

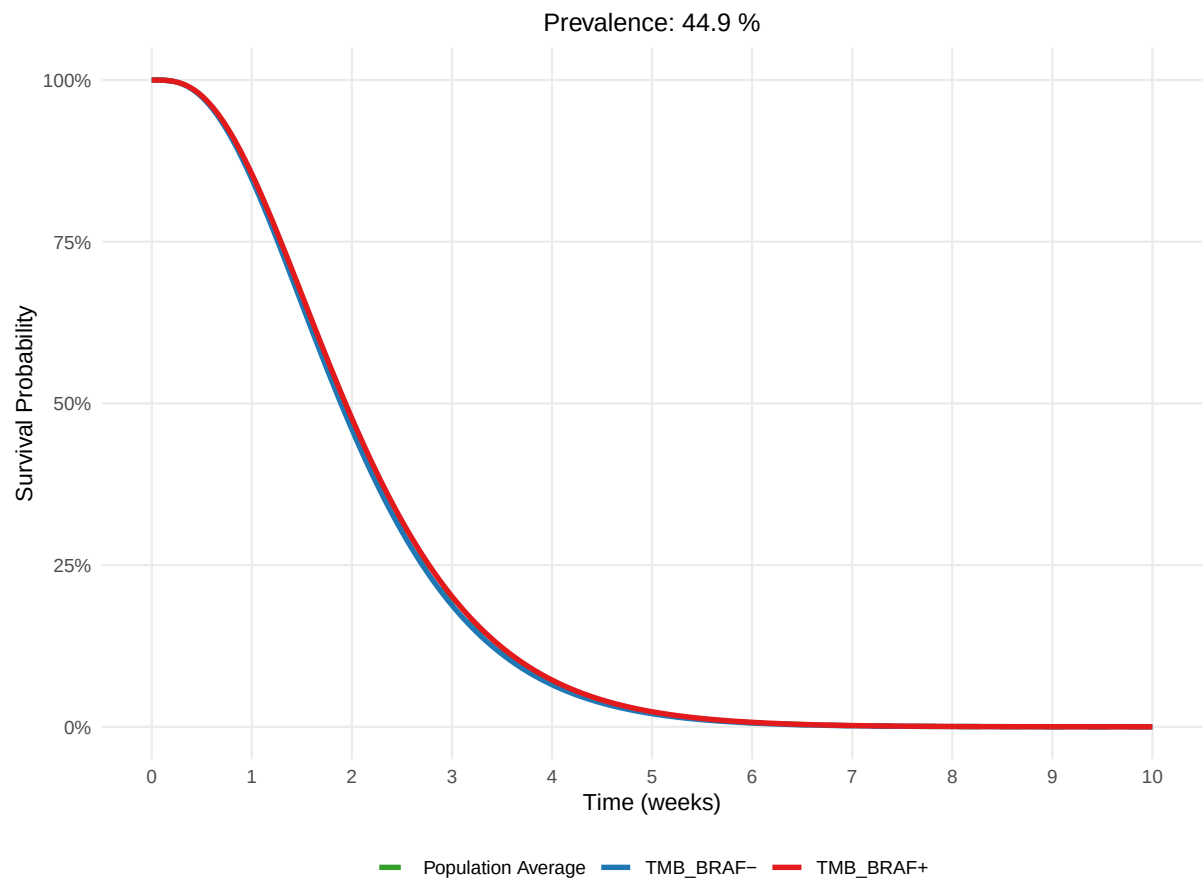


Figure 15: Overall survival for TMB_BRAF biomarker-positive versus biomarker-negative patients

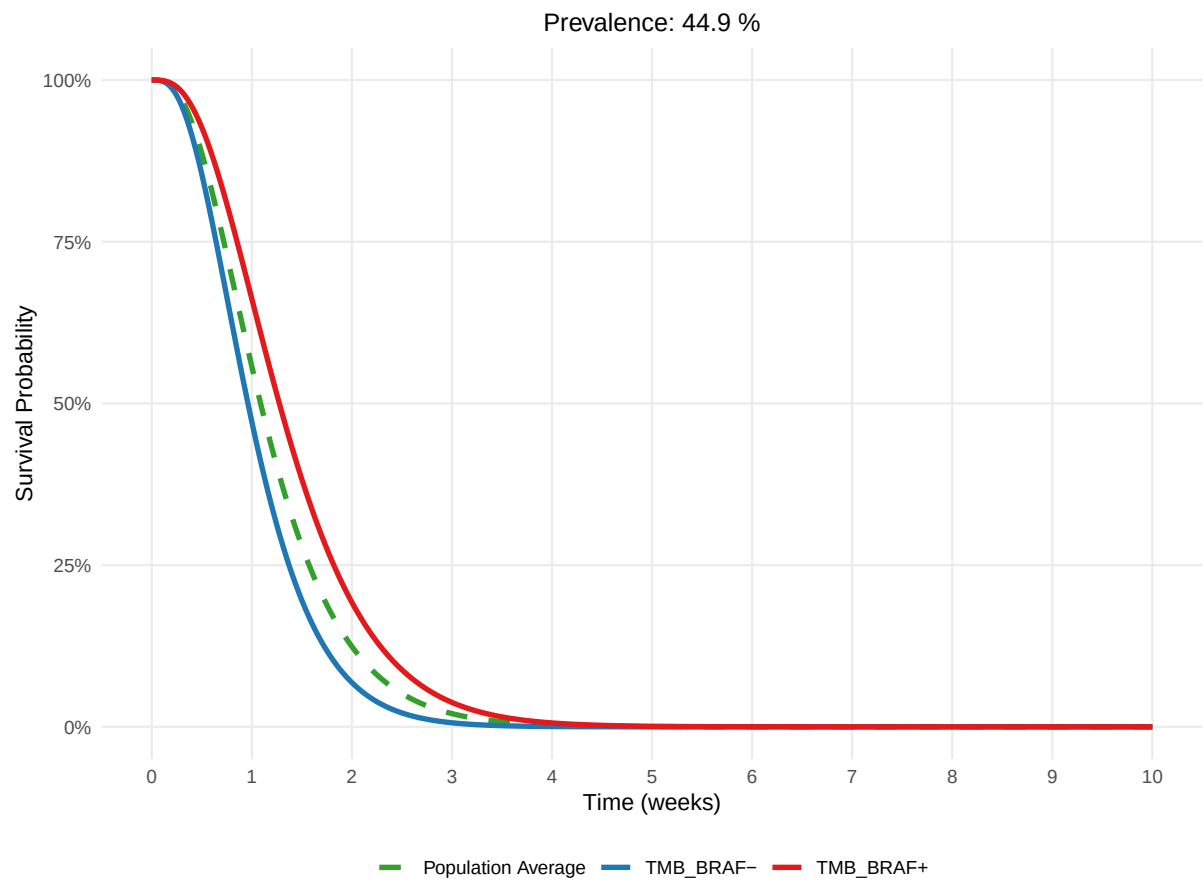


Figure 16: Progression-free survival for TMB_BRAF biomarker-positive versus biomarker-negative patients

8 Summary

- Parametric survival models were fitted for both overall survival (OS) and progression-free survival (PFS) outcomes
- Model selection was based on Akaike Information Criterion (AIC) to optimize predictive performance
- Both full models (including age and sex adjustment) and models without age/sex adjustment were evaluated
- Biomarker-specific survival patterns were characterized for CRP, TLR, and TMB_BRAF strategies
- Visual validation against Kaplan-Meier estimates confirms good model fit across all tested distributions

Report completed on: 2025-11-03

Repository: [ben-geisler/METIMMOX-1](#)

Report version: 1.0